UNIT – V

Plastic Analysis and Design

Syllabus: Introduction to Plastic Analysis – Theory of Plastic Analysis – Design of Continuous Beams and Portal Frames Using Plastic Design Approach.

What's inside these notes

- Elastic vs plastic behaviour, ductility, assumptions
- Plastic moment M_p, plastic section modulus, shape factor
- Plastic hinge, collapse mechanisms, load factor
- Theorems: static, kinematic (upper/lower bound), uniqueness
- Methods: virtual work (mechanism) & statical method
- Plastic design of continuous beams (examples)
- Plastic design of portal frames – beam, sway & combined mechanisms

Exam tip: Plastic collapse problems: (1) find hinge positions, (2) write virtual work equation for EACH mechanism, (3) the LEAST collapse load (or the GREATEST M_p required) governs, (4) check M ≤ M_p everywhere.

Codes used: IS 800:2007 (Limit State Method), IS 808 / SP-6(1) steel tables, IS 875 (loads), IS 2062 (steel grades). Take E = 2×10^5 MPa, f_y = 250 MPa, f_u = 410 MPa unless stated.

1. Introduction to Plastic Analysis

Elastic design limits stress at the extreme fibre to f_y (first yield). But ductile steel can carry more load: after the extreme fibres yield, the **yielded zone spreads inwards**, finally the whole section is plastic and the structure **collapses** only after enough **plastic hinges** form to make a **mechanism**.

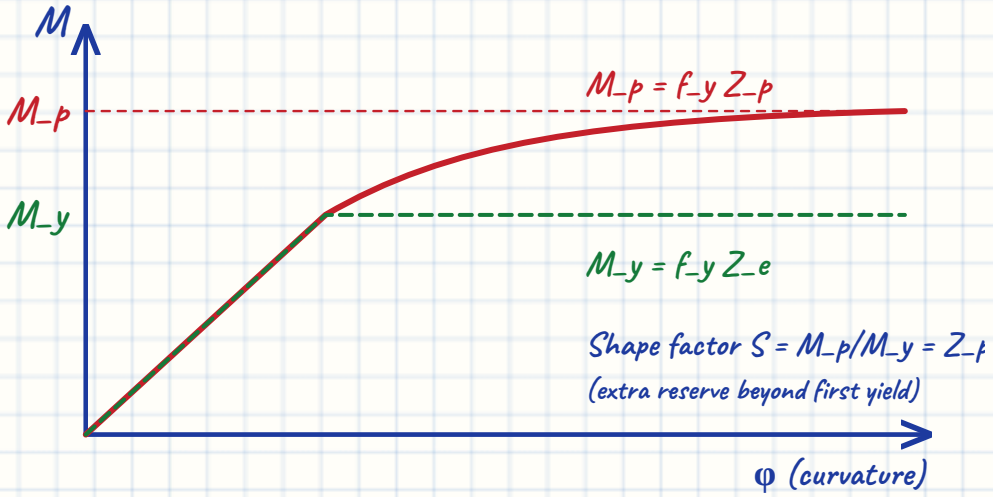
Plastic analysis finds the **collapse load (ultimate load)** directly, uses the full strength reserve and gives a more rational **load factor** $\lambda = W_c$ / W_working.

Why plastic analysis?

- More **economical** (10–30 % saving, esp. in continuous beams/frames)
- Gives the **true load factor** against collapse; redistribution of moments is accounted for
- Simple (statics only, no compatibility/ stiffness needed)
- Applicable to **ductile** steel (E250 ... E450) with **plastic or compact sections**.

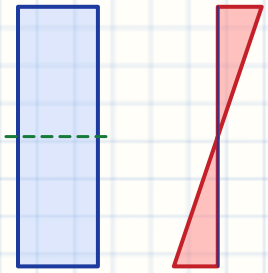
Assumptions

- Stress–strain curve is **idealised** elastic–perfectly plastic ($\sigma = f_y$ after yield) ; strain hardening ignored.
- **Plane sections remain plane** after bending ; tension = compression behaviour.
- Material is **homogeneous & isotropic** ; ductile (large strain without fracture)
- **Axial & shear effects** on M_p neglected (or corrected) ; local & lateral buckling prevented (adequate bracing, compact sections) ; stable under deflection (**small deformation** theory) ; loads are **proportional & static** (increase together).
- Connections are **full strength** & able to transmit M_p.



Moment–curvature relation (idealised)

2. Plastic Moment & Shape Factor



Stress distribution as the moment increases

- **Elastic** : $f = M\,y / I$; extreme fibre = f_y at **yield moment** $M_y = f_y\,Z_e$
- **Elasto-plastic** : yielded zones spread in from the extreme fibres (elastic core remains)
- **Fully plastic** : all fibres at f_y (tension below, compression above) ⇒ **plastic moment** (limit of capacity)

$$M_p = f_y\,Z_p\,,\; M_y = f_y\,Z_e$$

Plastic neutral axis

At full plasticity the **N.A. divides the area into two EQUAL halves** (so that C = T) — the **equal-area axis** (differs from elastic centroidal axis for un-symmetric sections).

$$Z_p = (A/2)(\bar{y}_1 + \bar{y}_2) = A_c\,y_c + A_t\,y_t$$ (y = distance of centroid of each half-area from the equal-area axis)

Shape factor

$$S = M_p / M_y = Z_p / Z_e$$

Section	S
Rectangle	**1.50**
Solid circle	1.70
Diamond (square on diagonal)	2.00
Triangle	2.34
I-section / channel (rolled)	**1.10 – 1.20** (≈1.14)
Thin circular tube	1.27
T-section	1.8 (approx.)

Higher S → more reserve beyond yield. For I-sections the flanges give most Z_e, so S is small.

EXAMPLE 1 – RECTANGLE B × D

$Z_e = bd^2/6$; $Z_p = 2 \times (bd/2) \times (d/4) = bd^2/4$ ⇒ **$S = (bd^2/4)/(bd^2/6) = 1.5$** ; $M_p = f_y\,b\,d^2/4$.

Example 2 – Built-up I-section : flanges 150×20, web 10×300 (total depth 340)

$I_z = [150\times340^3 - 140\times300^3]/12 = [5.8956\times10^9 - 3.78\times10^9]/12 = 1.763\times10^8\ mm^4$; $Z_e = I/170 = \mathbf{1.037\times10^6\ mm^3}$

Equal-area axis = centroidal axis (symmetric) → $Z_p = 2[150\times20\times160] + 2[10\times150\times75] = 960\,000 + 225\,000 = \mathbf{1.185\times10^6\ mm^3}$

$$S = 1.185/1.037 = 1.14$$

$M_p = 250 \times 1.185\times10^6 = \mathbf{296\ kN\cdot m}$; $M_y = 259\ kN\cdot m$

Example 3 – T-section (equal-area axis)

T : flange 100 × 10 (top), web 10 × 90 (total depth 100) ; $A = 1000 + 900 = 1900\ mm^2$ → half = $950\ mm^2$.

Flange area 1000 > 950 ⇒ **plastic N.A. lies inside the flange**, at $y = 950/100 = \mathbf{9.5\ mm}$ **from the top** (the elastic N.A. is at 28.7 mm – they differ!).

$Z_p = \Sigma\, A\, y$ (each part about the plastic axis) $= 950 \times 4.75 + 50 \times 0.25 + 900 \times 45.5 = 4512 + 12 + 40\,950 = \mathbf{45\,475\ mm^3}$

Elastic : $\bar{y} = (1000\times5 + 900\times55)/1900 = 28.7\ mm$; $I = 1.80\times10^6\ mm^4$; $Z_e = I/71.3 = 25\,240\ mm^3$

$$S = 45\,475/25\,240 = 1.80$$

Example 4 – Statical method for the propped cantilever (UDL w)

Moment at fixed end A : $M_A = -M_p$ ⇒ taking moments about A : $R_B\, L - wL^2/2 = -M_p$ ⇒ $\mathbf{R_B = wL/2 - M_p/L}$.

Sagging moment at distance x from the prop B : $M = R_B\, x - w\, x^2/2$; maximum where shear = 0 ⇒ $\mathbf{x = R_B/w}$, $M_{max} = R_B^2/(2w)$.

Plastic condition $M_{max} = M_p$ ⇒ $(wL/2 - M_p/L)^2 = 2\, w\, M_p$. Let $k = wL^2/M_p$: $(k/2 - 1)^2 = 2k$ ⇒ $k^2 - 12k + 4 = 0$ ⇒ $\mathbf{k = 6 + \sqrt{32} = 11.66}$

$$w_c = 11.66\, M_p/L^2 \checkmark \text{ (same as the kinematic result) and } x = R_B/w = 0.414\, L$$
$$\text{from the prop} \Rightarrow \text{upper bound = lower bound = true collapse load.}$$

Example 5 – Shape factor of a solid circle (diameter d)

$Z_e = \pi\, d^3/32$; plastic N.A. is the diameter, each half-area $= \pi\, d^2/8$ with centroid at $2d/(3\pi)$ from the axis

$Z_p = 2 \times (\pi d^2/8) \times (2d/3\pi) = \mathbf{d^3/6}$ ⇒ $S = (d^3/6)/(\pi d^3/32) = 16/(3\pi) = \mathbf{1.70}$.

3. Plastic Hinge & Mechanisms

Plastic hinge = a section where $M = M_p$ – the section rotates (plastic rotation) at **constant moment M_p** like a hinge with a friction torque. Differences from a real hinge:

carries M_p ; **unidirectional** (rotation in direction of M only) ; has a small finite length in reality but is idealised at a point.

- **Collapse mechanism** : enough hinges form so that the structure becomes a **kinematic chain (mechanism)** – deflections grow without further load increase.
- Number of hinges required for complete collapse : **n = r + 1** (r = **degree of static indeterminacy**) for a **complete** collapse ; fewer hinges (**partial / over-complete collapse**) in some cases.



● = plastic hinge (r+1 hinges for collapse; r = degree of indeterminacy)

Collapse mechanisms of beams

Locations of plastic hinges

- Points of **maximum moment** : under concentrated loads, fixed/continuous supports, at joints of frames (corners), within spans for UDL (where shear = 0).
- Hinge at a **joint of 3 or more members** forms in the weakest member.

Load factor

λ (load factor) = Collapse load W_c / Working load W

IS 800:2007 works with **factored loads** (Table 4, e.g. 1.5 (DL + LL)) and design $M_d = Z_p f_y/\gamma_{m0}$ (older IS 800:1984 used a load factor of **1.7** for DL + LL, 1.3–1.4 with wind).

Mechanisms in frames

- **Beam mechanism** – hinges in beam (3 hinges)
- **Panel / sway mechanism** – columns sway (4 hinges: tops & bottoms of columns)
- **Gable mechanism** – rafter / ridge of gable frames
- **Joint mechanism** – rotation at joint of 3+ members (rarely)
- **Combined mechanism** – combination of independent mechanisms to eliminate a hinge → **lowest collapse load**.

Number of independent mechanisms n = N – r (N = number of possible hinge locations ; r = degree of redundancy).

4. Theorems of Plastic Collapse

For a proportionally increasing load with a given structure & M_p :

- **Static (Lower bound) theorem** – A load W_s for which a **statically admissible moment diagram** exists (equilibrium + $M \leq M_p$ everywhere) is **≤ actual collapse load** (**safe** estimate).

- **Kinematic (Upper bound) theorem** – A load W_k computed for **any assumed mechanism** (by virtual work) is **≥ actual collapse load** (**unsafe** estimate).
- **Uniqueness theorem** – The collapse load W_c is the one for which BOTH are satisfied : the moment diagram is statically admissible **and** the mechanism is formed ($W_s = W_k$).

$$W_s \le W_c \le W_k$$

So **try all possible mechanisms** → collapse load = **smallest** W_k; then verify the **B.M. diagram** (statical check) to make sure no $|M| > M_p$.

Methods of analysis

- **Kinematic (mechanism) method** – virtual work: **External work = Internal work**

$$\Sigma W\delta = \Sigma M_p \theta$$

δ = virtual displacement at load, θ = virtual hinge rotation. Internal work is always positive : $M_p \times |\theta|$. At a hinge between two members rotating by θ_1, θ_2 : relative rotation.

- **Statical method** – draw free + reactant B.M.D. : assume redundant, put $M = \pm M_p$ at the hinges and use equilibrium to find W.
- **Moment-balancing / Instantaneous centre method** (frames).

5. Collapse Load of Beams (Standard Results)

Let M_p = plastic moment, span L. **Virtual work** : $W \times \delta = M_p \Sigma \theta$.

(a) Simply supported, central point load W

Hinge at centre; $\delta = (L/2)\ \theta$; rotation at hinge = 2θ : $W\ (L/2)\ \theta = M_p\ (2\theta)$ →

$$W_c = 4\ M_p / L \text{ (UDL : } w_c = 8\ M_p / L^2)$$

(b) Fixed beam, central point load

Hinges at both ends and centre : rotations θ, 2θ, θ (total 4θ) : $W\ (L/2)\ \theta = 4\ M_p\ \theta \Rightarrow$ **$W_c = 8\ M_p / L$**

UDL : work = $w \times (½ \cdot L \cdot L\theta/2) = w\ L^2\ \theta/4 = 4\ M_p\ \theta \Rightarrow$ **$w_c = 16\ M_p / L^2$**

(c) Propped cantilever (fixed A, prop at B), central point load

Hinges at A and under the load : rotations θ and 2θ (total 3θ) → $W\ (L/2)\ \theta = 3\ M_p\ \theta \Rightarrow$ **$W_c = 6\ M_p/L$**

UDL : hinge at A and a sagging hinge at **0.586 L from A (= 0.414 L from the prop)** →

$$w_c = 11.66\ M_p / L^2\ (= 2(3 + 2\sqrt{2})\ M_p/L^2)$$

(d) Cantilever, end load : $W_c = M_p / L$.

Beam	Hinges	W_c (point at centre)	w_c (UDL)
Simply supported	1	$4M_p/L$	$8M_p/L^2$
Fixed–fixed	3	$8M_p/L$	$16M_p/L^2$
Propped cantilever	2	$6M_p/L$	$11.66M_p/L^2$
Cantilever (end load)	1	M_p/L	—

Example 6 – Propped cantilever with UDL (derivation)

Fixed end A, prop B. Hinges at A and at C, a distance x from A. Let AC rotate θ about A, CB rotate φ about B ; deflection at C : $\delta = x\theta = (L - x)\varphi \Rightarrow \varphi = x\theta/(L - x)$.

Hinge rotations : $A = \theta$; $C = \theta + \varphi = \theta\, L/(L - x)$; $B = 0$ (pin).

Internal work $= M_p\,[\theta + \theta\, L/(L - x)] = M_p\,\theta\,(2L - x)/(L - x)$

External work $= w \times \tfrac{1}{2}\, L\, \delta = \tfrac{1}{2}\, w\, L \times \theta$

$$w = 2\,M_p\,(2L - x)\,/\,[L \times (L - x)]$$

Minimise w.r.t. x : $x^2 - 4Lx + 2L^2 = 0 \Rightarrow$ **$x = (2 - \sqrt{2})L = 0.586\,L \Rightarrow w_c = 11.66\,M_p/L^2$**

6. Plastic Design of Continuous Beams

Idea: For the **factored loads**, find the collapse mechanism of each span and the required M_p ; select a section with $Z_p \geq M_p\,\gamma_{m0}/f_y$ (IS 800: $M_d = Z_p\, f_y/\gamma_{m0} \geq M_u$).

Possible mechanisms in a continuous beam

- Each span can fail **independently** (a span between two hogging hinges + one sagging hinge = **fixed-beam mechanism**) ; end spans (propped-cantilever type) ; **combined** mechanisms are not possible across supports (no relative displacement at supports).
- For a **uniform M_p**, consider every span separately & take the **largest required M_p**.

Steps

- 1. Factored loads on each span (1.5 (DL + LL) or the relevant IS 800 Table 4 combination).
- 2. Choose mechanism for each span : **end span** with simple support → **propped-cantilever** (hinge near 0.414 L from the outer support : $w\,M_p = 11.66\ldots$), **interior span** both ends continuous → **fixed-beam type** ($w = 16\,M_p/L^2$).
- 3. Compute $M_p(\text{req}) = W\,L\,/\,k$ for each span. Take **max** ; if different sections per span economical use span-wise M_p (but hinges at supports use the **smaller** M_p of the two adjacent spans).
- 4. **Select $Z_p = M_p\,/\,(f_y/\gamma_{m0})$** ; check shape factor/class (plastic), shear $V \leq 0.6\,V_d$, web buckling, **lateral bracing**, deflection at working load.
- 5. **Check the B.M. diagram** : $|M| \leq M_p$ everywhere.

Example 7 – Two-span continuous beam, equal spans L = 6 m, UDL w_u = 40 kN/m

FACTORED

Each span is like a propped cantilever (hinge over middle support + sag hinge at 0.414 L from the outer support... both spans identical).

$$w_u = 11.66\, M_p/L^2 \Rightarrow M_p = 40 \times 6^2/11.66 = 123.5\ kN\cdot m$$

$Z_p(req) = 123.5\times10^6 \times 1.1/250 = 543\,600\ mm^3 =$ **544 cm³** → ISMB 300 ($Z_p \approx 654\ cm^3$) ✓ ; ISMB 250 ($Z_p \approx 467$) ×.

Check shear : $R_A = wL/2 - M_p/L = 120 - 20.6 = 99.4$ kN ; total at inner support = $2(240 - 99.4) = 281$ kN ⇒ V_u (each side) = **140.6 kN** < $0.6\ V_d$ (≈ 182 kN for ISMB 300) ✓

Provide ISMB 300 (with lateral bracing for compression flange near supports).

EXAMPLE 8 – CONTINUOUS BEAM, SPANS 5 M AND 6 M WITH DIFFERENT LOADING (OUTLINE)

Span AB = 5 m, central point load W_1 = 100 kN (factored) ; BC = 6 m, UDL w = 30 kN/m (factored), A, C simply supported.

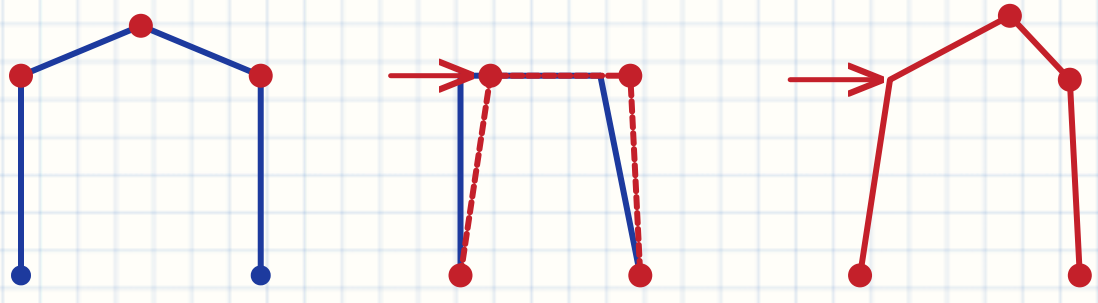
Mechanism AB (propped cantilever with point load at centre) : $W_1 = 6\ M_p/L \Rightarrow M_p = 100\times5/6 =$ **83.3 kN·m**

Mechanism BC (propped cantilever with UDL) : $w = 11.66\ M_p/L^2 \Rightarrow M_p = 30 \times 36/11.66 =$ **92.6 kN·m** (governs)

Hence **M_p = 92.6 kN·m → Z_p = 407 cm³ ⇒ ISMB 250** ($Z_p \approx 467\ cm^3 \geq 407$) and check the B.M.D. (no moment above M_p).

7. Plastic Analysis of Portal Frames

Fixed-base single-bay portal frame (columns h, beam L, **uniform M_p**, vertical load W at mid-span of beam, horizontal H at eaves level). Possible plastic hinge positions : **A, E (bases), B, D (eaves), C (under W)** → **N = 5**, r = 3 ⇒ **independent mechanisms n = 5 – 3 = 2** (beam, sway) and **1 combined**.



Beam, sway & combined mechanisms of a portal frame

Virtual work equations (θ = virtual rotation)

- **Beam mechanism** : hinges at B, C, D (θ, 2θ, θ) ⇒ $4M_p\theta = W\ (L/2)\ \theta \Rightarrow W = 8\ M_p / L$

- **Sway mechanism** : hinges at A, B, D, E (θ each), eaves displace hθ ⇒ 4M_pθ = H h θ ⇒ **H = 4 M_p / h**
- **Combined mechanism** : eliminate hinge at B (rotations θ and −θ cancel) : hinges A(θ), C(2θ), D(2θ), E(θ) → 6M_pθ

$$6\,M_p = H\,h + W\,L/2$$

(Hinge at D rotates θ + θ = 2θ, because the sway rotation and the beam rotation add.)

> With only vertical loads the combined and sway mechanisms vanish → **beam mechanism**. With a fixed-base frame, pinned bases give hinge at A, E removed (3 hinges in sway; M_p required increases).

Procedure : compute M_p from each mechanism → **maximum M_p (least collapse load)** governs → check the **moment diagram** (equilibrium) and **column design for axial + M_p**.

EXAMPLE 9 – FIXED-BASE PORTAL FRAME : H = 4 M, L = 6 M, W = 100 KN (MID-SPAN), H = 50 KN (EAVES) – FACTORED

Beam : W (L/2) θ = 4 M_p θ ⇒ M_p = W L / 8 = 100 × 6 / 8 = **75 kN·m**

Sway : H h = 4 M_p ⇒ M_p = 50 × 4 / 4 = **50 kN·m**

Combined : 6 M_p = H h + W L/2 = 50 × 4 + 100 × 3 = 500 ⇒ M_p = **83.3 kN·m** ← largest

Required **M_p = 83.3 kN·m** (combined mechanism governs – the **least collapse load**).

Z_p(req) = 83.3×10⁶ × 1.1/250 = **367 cm³** → choose ISMB 250 (Z_p ≈ 1.14 × 410 = 467 cm³) or equivalent.

Moment check (statics) : with hinges A, C, D, E at ±M_p, equilibrium of the beam and columns gives M_B (the hinge eliminated in the combined mechanism) — verify $|M_B| \le M_p$; if so, upper bound = lower bound and this is the true collapse mechanism.

Effect of axial force & shear on M_p

- Axial load N reduces plastic moment : **$M_{pc} = M_p\,[1 - (N/N_y)^2]$** (rectangle) ; for I-section, if $N \le 0.15\,N_y$ neglect, else use reduced M_{pc} (see IS 800 Cl. 9.3).
- High shear ($V > 0.6\,V_d$) reduces M_p (see Unit III).
- Columns also checked for **buckling** with plastic M_p; ensure **adequate bracing, stiff joints**, and **rotation capacity** (plastic or compact sections only).

8. Gable Frames, Instantaneous Centres & Checks

Gable frame mechanism

Rafters inclined; hinges at eaves, ridge (apex) and bases. Use **instantaneous centres (I.C.)** of each rigid link to relate rotations and displacements.

- Beam (rafter pair) mechanism: rafter rotates about I.C. ; sway : eaves displace horizontally; combined etc.
- Virtual work with horizontal & vertical displacement of load points: δ_H = (distance of load point from I.C.) × θ etc.

Moment-balance (statical) check of portal frame

After finding M_p from the critical mechanism, check:

- 1. Draw free B.M.D. (as for simply supported beam) and reactant diagram from M_A, M_E, M_B, M_D... with chosen redundants (H_A, M_A).
- 2. Superpose ; verify **$|M| \le M_p$ everywhere**. If any $M > M_p$, another mechanism governs.
- 3. Equilibrium equations ΣM, ΣH, ΣV hold.

Advantages / limitations of plastic design

Advantages	Limitations
Economical, uses reserve strength	Only for ductile steel & **compact sections**
Simple analysis (statics)	Deflection & serviceability must be checked separately (working loads)
True load factor	Local/lateral buckling, fatigue & fracture must be controlled
Redistribution captured	Repeated / alternating loads may cause incremental collapse

EXAMPLE 10 – FIXED BEAM, L = 8 M, UDL w_u = 30 KN/M (FACTORED)

$w_c = 16\, M_p/L^2 \Rightarrow$ **$M_p = 30 \times 8^2/16 = 120$ kN·m** (hogging at both ends & sagging at centre, each = M_p).

$Z_p(req) = 120\times10^6 \times 1.1/250 =$ **528 cm³** → ISMB 300 ($Z_p \approx 654$) ✓.

(Elastic design needs $M = wL^2/12 = 160$ kN·m at the supports $\Rightarrow Z_e = 160\times10^6\times1.1/250 = 704$ cm³ ⇒ ISMB 350 – plastic design saves one size.)

EXAMPLE 11 – SAME PORTAL FRAME BUT WITH PINNED BASES (H = 4, L = 6, W = 100, H = 50)

Hinge positions : B, C, D only ($N = 3$, $r = 1 \Rightarrow n = 2$ mechanisms).

Beam : $4\, M_p = W\, L/2 \Rightarrow$ **M_p = 75 kN·m**

Sway : hinges at B, D only : $2\, M_p\, \theta = H\, h\, \theta \Rightarrow M_p = 50\times4/2 =$ **100 kN·m**

Combined : eliminate B ; C (2θ), D (2θ) : $4\, M_p\, \theta = H\, h\, \theta + W\, (L/2)\, \theta \Rightarrow M_p = (200 + 300)/4 =$ **125 kN·m** ← governs

Pinned bases need M_p = 125 against 83.3 for fixed bases (≈ 50 % more steel in the frame; foundations are cheaper, though).

9. Rotation Capacity, Redistribution & Practical Points

- **Rotation capacity** $R = (\theta_u - \theta_p)/\theta_p$: ability of the hinge to rotate at M_p without losing strength. It needs **plastic (Class 1) sections** ; limits : flange ≤ 9.4 ε, web ≤ 84 ε (rolled I) – see Unit I.
- **Moment redistribution** : in an indeterminate structure, when the first hinge forms, extra load is carried by the remaining sections until the mechanism is complete – this is the extra strength plastic design uses.

- **Lateral bracing** : braces close to hinge locations so LTB does not occur before the hinge rotates.
- **Connections** must be **stronger** than the members (full-strength, haunches or stiffeners at knees).
- **Incremental collapse (ratcheting)** & **alternating plasticity** under repeated loads – shakedown theory (awareness only).
- **Deflection** is checked at **working** (unfactored) loads – elastic.

EXAMPLE 12 – THREE-SPAN CONTINUOUS BEAM, EQUAL SPANS 5 M, UDL 36 KN/M (FACTORED)

End spans (propped-cantilever type) : $M_p = w\,L^2/11.66 = 36 \times 25/11.66 =$ **77.2 kN·m**

Interior span (both ends continuous – fixed-beam type) : $M_p = w\,L^2/16 = 36 \times 25/16 =$ **56.3 kN·m**

Uniform section ⇒ take the **larger** : $M_p = 77.2$ kN·m ⇒ $Z_p = 77.2\times10^6 \times 1.1/250 =$ **340 cm³** → ISMB 250 ($Z_p \approx 467$) ✓.

(The interior span is under-stressed – its sagging hinge does not form first; hogging hinges at supports are shared.)

Plastic section modulus of a doubly symmetric I-section (formula)

$$Z_p = b\,t_f\,(d - t_f) + t_w\,(d - 2\,t_f)^2 / 4 \quad (d = \text{overall depth},\ b = \text{flange width})$$

Check ISMB 300 : $140\times13.1\times286.9 + 7.7\times273.8^2/4 = 526\,185 + 144\,310 \approx 670$ cm³ (tables give ≈ 654 cm³ because of root fillets/tapered flanges – always use SP-6 values).

Effect of axial force on M_p (columns of frames)

- For a **rectangle** : $M_{pc} = M_p\,[1 - (N/N_y)^2]$, $N_y = A\,f_y$.
- For **I-sections** (major axis) the reduction is small : $M_{pc} \approx 1.18\,M_p\,(1 - N/N_y) \le M_p$; **neglect** it when $N \le 0.15\,N_y$.
- Columns of frames are checked for **combined axial + bending** and **buckling** after M_p is fixed.

Elastic design vs plastic design

Feature	Elastic (WSD)	Plastic
Basis	first yield f_y at extreme fibre	formation of mechanism
Capacity	$M_y = f_y\,Z_e$	$M_p = f_y\,Z_p$ (≈ 14 % more for I)
Analysis	indeterminate (compatibility)	statics + virtual work
Safety measure	factor of safety on stress	load factor on collapse
Economy	less	10–30 % saving (continuous beams/frames)
Applicable to	any steel	ductile steel, compact sections

> Exam trick: in portal-frame questions always write **three equations** (beam, sway, combined) – then state **the greatest M_p** and say "combined mechanism governs, so M_p = … ".

Important formulae – Unit V

$M_p = f_y Z_p$; $M_y = f_y Z_e$; $S = Z_p/Z_e$ ($I \approx 1.14$, rectangle 1.5, circle 1.7)

SS beam : $W_c = 4M_p/L$; $w_c = 8M_p/L^2$

Fixed beam : $W_c = 8M_p/L$; $w_c = 16M_p/L^2$

Propped cantilever : $W_c = 6M_p/L$; $w_c = 11.66M_p/L^2$

Portal : beam $4M_p\theta = W(L/2)\theta$; sway $4M_p = Hh$; combined $6M_p = Hh + WL/2$

Hinges for collapse = $r + 1$; independent mechanisms = $N - r$; $\Sigma W\delta = \Sigma M_p\theta$

Important exam questions

1. Define plastic hinge, shape factor, load factor; state assumptions of plastic theory.
2. Derive M_p and S for rectangular & I-section.
3. State and prove the static, kinematic and uniqueness theorems.
4. Find the collapse load of a propped cantilever / fixed beam with UDL.
5. Design a two-span continuous beam by plastic theory.
6. Analyse a fixed-base portal frame – beam, sway, combined mechanisms ; select the section.